

AISI 1080 Steel, as rolled**Categories:** [Metal](#); [Ferrous Metal](#); [Carbon Steel](#); [AISI 1000 Series Steel](#); [High Carbon Steel](#)**Material Notes:** Similar to AISI 1055 and AISI 1074.**Key Words:** UNS G10800, carbon steels**Vendors:** No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
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Density	7.85 g/cc	0.284 lb/in ³	
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Mechanical Properties	Metric	English	Comments
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Hardness, Brinell	293	293	
Hardness, Knoop	319	319	Converted from Brinell
Hardness, Rockwell B	99	99	Converted from Brinell
Hardness, Rockwell C	31	31	Converted from Brinell
Hardness, Vickers	309	309	Converted from Brinell
Tensile Strength, Ultimate	965 MPa	140000 psi	
Tensile Strength, Yield	585 MPa	84800 psi	
Elongation at Break	12 %	12 %	In 50 mm
Reduction of Area	17 %	17 %	
Modulus of Elasticity	205 GPa	29700 ksi	Typical for steel
Bulk Modulus	160 GPa	23200 ksi	Typical for steel
Poissons Ratio	0.29	0.29	Typical For Steel
Shear Modulus	80.0 GPa	11600 ksi	Typical for steel
Izod Impact	7.00 J	5.16 ft-lb	

Electrical Properties	Metric	English	Comments
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Electrical Resistivity	0.0000180 ohm-cm @Temperature 20.0 °C	0.0000180 ohm-cm @Temperature 68.0 °F	condition of specimen unknown
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Thermal Properties	Metric	English	Comments
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CTE, linear 	11.0 µm/m-°C @Temperature 0.000 - 100 °C	6.11 µin/in-°F @Temperature 32.0 - 212 °F	
	12.2 µm/m-°C @Temperature 0.000 - 300 °C	6.78 µin/in-°F @Temperature 32.0 - 572 °F	
	13.7 µm/m-°C @Temperature 0.000 - 500 °C	7.61 µin/in-°F @Temperature 32.0 - 932 °F	
Specific Heat Capacity	0.490 J/g-°C	0.117 BTU/lb-°F	50-100°C
Thermal Conductivity	47.7 W/m-K	331 BTU-in/hr-ft ² -°F	0°C

Component Elements Properties	Metric	English	Comments
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Carbon, C	0.75 - 0.88 %	0.75 - 0.88 %	
Iron, Fe	98.13 - 98.65 %	98.13 - 98.65 %	As remainder
Manganese, Mn	0.60 - 0.90 %	0.60 - 0.90 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Sulfur, S	<= 0.050 %	<= 0.050 %	

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.